University of Birmingham
School of Engineering
Department of Mechanical Engineering

Advanced Project

04 25887
(60 Credits Summer Project)

G U I D E L I N E S
2025-26

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Content **Page No.**

1. Module Learning Outcomes

The following are the generic learning outcomes of Advanced Project module:

- Use a combination of general and specialist engineering knowledge and understanding to optimise the application of an existing or emerging technology appropriate to the project.
- Apply appropriate theoretical and practical methods to the analysis and solution of an engineering problem including defining a suitable project, design and conduct appropriate research make appropriate recommendations and learn from feedback.
- Plan for effective project implementation and take responsibility for the effective management of shared resources and time according to an agreed work plan which is reviewed during the course of the project.
- Demonstrate effective interpersonal skills including effective written and verbal communication on complex technical material and with others in the working environment.
- Demonstrate a personal commitment to professional standards, recognising obligations to society, the profession and the environment including complying with University rules of conduct, health, safety and welfare issues, sustainability, training and ethical requirements appropriate to the project.
- Demonstrate detailed use and critical assessment of technical literature and other information sources.
- Provide clear and relevant recommendations and conclusions supported by evidence.
- Work independently, managing personal work programme to meet project milestones.
- Provide critical and reflective assessment of project findings.

2. Working on Your Project

The project period starts in the first week of June and ends in the first week of September. The module description document specifies that you should spend 584 hours of independent study on your project over the project period. It is your responsibility to arrange regular meetings with your project Supervisor and discuss all relevant aspects of your project with them during these meetings.

You are expected to show initiative and to attempt to solve problems that occur during the course of the project. You should not expect your project Supervisor to give you detailed instructions about what to do; the Supervisor's role is to provide guidance.

3. Risk Assessment and Health and Safety

The **Risk Assessment Form** (Available on Canvas) must be completed in order to identify real risks and responsibilities. Therefore, you should not invent trivial risks but should concentrate on things that could really endanger yourself and/or others. You are responsible for your own actions but, in most cases, are not responsible for the maintenance of the equipment that you use; your risk assessment should be realistic about your own responsibilities. For example, if equipment is faulty or apparently dangerous, your responsibility is to report the problem to a responsible person and stop using the equipment until the problem is rectified. To be signposted for advice on risk assessment please contact your Supervisor in the fist place.

4. Ethical Review of the Project

In all cases, an **Ethical Review Form** (Available on Canvas) must be completed (plus a Full Ethical Review and/or Information and Consent form if then deemed necessary). Electronic copies of these (with the exception of any Information and Consent forms, copies of which should simply be retained by you) should be emailed to your Supervisor as soon as the project starts. Failure to comply could result in the project being failed. Consult with your supervisor if you need help completing this form.

5. Regular Meetings with Your Supervisor (Project Meeting Form)

You must arrange to hold regular meetings (face-to-face or via video conferencing if necessary) with your supervisor, or a member of the supervision team. It is your responsibility to contact your supervisor and request for meetings. A record of these meetings should be made by completing and submitting to Canvas a **Project Meeting Form** (available in Appendix D) on regular bases as required. Furthermore, you are expected to be professional in your conduct towards all matters related to your project and adopt an engaging attitude with determination to progress and succeed.

6. Assessment Information and Submission Deadlines

Your Assessment will include two separate elements: (i) A Video Presentation, and (ii) Final Report. Details of the report preparation and formatting are given in section 10 and Appendix A. For video presentation you are required to submit a **10-minute** (max.) **recorded PowerPoint presentation**. It is important that you practice before submitting the final version, and that you do not exceed the maximum 10 minutes allowed. The presentation should be logically structured using a reasonable number of clear and high-quality slides. The first slide should include your full name, ID, your Supervisor's name and the title of your project. The main body of the presentation should include the required information given in Section 7. Clear and effective communication to convey key information using impactful and logically structured and organised slides with engaging narrative and images/figures constitute a high-quality presentation.

Assessment Element	Submission Deadline	Max. Mark
Video Presentation* (Submission to Canvas)	12 Noon Friday 26th June 2026	20%
Final Report Submission* (Submission to Canvas)	12 Noon Friday 4th September 2026	80%

**If your submission is not received on time, your final project mark may be reduced by up to 5% for each day (or part day) that it is late. For further information see "Code of Practice - Taught Programme and Module Assessment and Feedback", Section 4.2.4: Penalties for Late Submission of Work: https://intranet.birmingham.ac.uk/student/academic-support/registry/legislation/documents/public/cohort-legislation-2025-26/cop-taught-prog-assessment-25-26.pdf*

7. Video Presentation - Assessment Criteria

Your should provide the following information in your video presentation

Project Title
Provide a succinct and focused project title
Project Aim and Objectives
Aim - Describe the overall, but focused, aim of this project. Objectives - Give a bullet point description of the key Objectives of your research. Make sure each bullet point sufficiently explains the objective.
Problem Statement
Explain what problem you are trying to solve. Give a background and context to the problem. What are the industrial, commercial, environmental, and/or societal motivations for your project? What are the expected constraints and limitations that your project could be facing?
Current State-of-the-Art
Outline and describe (with references) the latest advancements in the relevant field(s) of your project
Approach and Methodology
How do you intend to solve the problem? What are the technical tasks? Describe the technical/scientific techniques/methods you are going to use to achieve the project's aim? How are you going to implement the techniques/methods to solve the problem?
Technical Milestones
What are the key Technical Milestones of your project? List and describe them with justifications. (Note: Do NOT include project management milestones such as submission deadlines, supervisor meetings, project write-up, etc.)
Expected Outputs and Deliverables
What will be the expected Technical Deliverables of your project? List and describe the nature of likely outputs. For example, will they be in the form of a novel design solution, an analytical analysis, a numerical analysis, simulation, etc.? List and clearly describe the project's expected outputs.
Project Impact
Describe the expected impacts and benefits of the project Note: Impact could be in the form of industrial/economic, commercial/business, environmental, societal What is the nature and range of the impacts? Who are impact stakeholders and how will they be implemented? What are the likely challenges and risks of the implementation of impacts?
Gantt Chart
Insert a technical time plan for the project duration (June/July/August 2025) in the form of a simple Gantt chart. Note: Do NOT include project management tasks such as submission deadlines, supervisor meetings, project write-up, etc.
References
List key references you have cited/used in this presentation Your references should mostly be from journal publications. Website references should be kept to a minimum. Use Vancouver referencing system.

Video Presentation *(Maximum 10 minutes duration)*	Max. Mark
Understanding of the problem Background Problem Statement Aim and Objectives Current State-of-the-Art Expected constraints/limitations	25
Proposed approach and expected outcomes Proposed Methodology Technical Milestones Technical Deliverables Impact Time plan/Gantt chart	25
Q&A Appropriate responses to the questions Effective oral communication	50
Total Presentation	**100**

(See Appendix B for Detailed Video Presentation Assessment Criteria)

8. Final Report - Assessment Criteria

Final Report	Max. Mark
Level of understanding of chosen topic (Background and Aim) - Demonstrate understanding of the topic in terms of depth and breadth. - Demonstrate evidence of independent study. - Demonstrate the ability to drawing upon the latest developments in the subject. - Demonstrate the ability to evaluate new knowledge. - Outline clear project aim and objectives	20
Quality of critical engagement with literature (Literature Review) - Evaluate the literature in the field of the research - Consolidate knowledge - Evaluate Key theories - Synthesise theories related to the research topics	20
Choice and justification of research methods or design methods (Methodology) - Adopt appropriate research methods and techniques to fit the research - Justify and evaluating the techniques used	20
Quality of analysis and evaluation of findings or design (Results) - Analyse and evaluating research findings - Synthesise between research findings and the literature review. - Demonstrate that the conclusions made are supported by the research.	20
Impact Statement - Identify relevant economic/industrial, commercial, environmental, and societal project impact(s) - Identify nature and range of impact(s) including benefits, implementation & stakeholder considerations	10

Standard of presentation & structure of work (Quality) - Producing appropriate written expressions in text and in figures and tables - Meeting all project milestones - Demonstrating a logical flow of the structure and navigation of work - Citing and listing references correctly	10
Total Report	**100**

(See Appendix C for Detailed Report Assessment Criteria)

9. Impact Statement

To help you engage more deeply with your project and understand the potential impact and implementation aspects of your project, you are required to prepare and submit a mandatory Impact Statement as part of your project. An 'impact statement' aims to describe the contribution of a body of research (or other project work) to the wider industrial/economic, environmental, and societal development.

Key questions to be considered are:

- WHAT type of benefit or impact is expected from the project
- WHO will benefit from the research?
- HOW will they benefit from the research?

A separate lecture on project impact will be organised in late June. All students are expected to attend. The lecture date/time/venue will be announced separately.

10. Final Report Submission Information

Use the following template on the Cover Page of the Final Report:

<table>
<tr><td colspan="2" align="center">University of Birmingham
College of Engineering and Physical Sciences

Department of Mechanical Engineering
School of Engineering

MSc Advanced Mechanical Engineering

Advanced Project

2025-26 Session

FINAL REPORT</td></tr>
<tr><td>Surname</td><td></td></tr>
<tr><td>First Name</td><td></td></tr>
<tr><td>ID number</td><td></td></tr>
<tr><td>Supervisor’s Name</td><td></td></tr>
<tr><td>Project Title</td><td></td></tr>
</table>

The *Final report* should be in a format suitable for submission as a technical paper to **Proceedings of the IMechE, Part C: Journal of Mechanical Engineering Science**; further details are given in the file IMechE Authors Guidelines. The main body of the text should not exceed 5000 words and contain no more than 10 figures (although a figure may consist of several related parts denoted a, b, c etc.) and 10 tables; for an experimental report, it will typically be divided into the following sections: Introduction, Methods, Results, Discussion, and Conclusions. Other section headings are allowable. Please put figures and tables at appropriate places in the text.

You should familiarise yourself with manuscript submission details and journal paper styles including method of referencing to understand how your work should be presented. Use the link below, then scroll halfway down the page and click “Submission Guidelines”, and then find “Manuscript Style” link:

https://journals.sagepub.com/author-instructions/PIC

Examples of recently published papers can be viewed through the link below:

NOTE: you should submit your report according to the manuscript submission guidelines mentioned above as the final published papers may look differently to the submitted format. There is no need to present the text in two columns.

Reports that are shorter and contain fewer figures and tables are acceptable provided they cover the project at the level of detail required; **your mark does not depend on how much you write or on the number of figures or tables**.

Further details on writing a report are given below. Your supervisor may ask for supplementary material to help in the assessment of your work; any such material should be submitted directly to him/her. The purpose of the report is to describe what you did in your project, present the results and to put your work into context.

You should avoid irrelevant material and **must not plagiarise** the work of others.

University's code of practice on academic integrity including plagiarism may be found here:

https://intranet.birmingham.ac.uk/student/academic-support/registry/legislation/documents/public/cohort-legislation-2025-26/cop-academic-integrity-25-26.pdf

If you have any doubts about what constitutes plagiarism, you should discuss them with your supervisor.

APPENDIX A – Final Report Format Guidelines

1. Introduction

The purpose of this document is to supplement the Instructions for Authors of Technical paper in Proceedings of the IMechE, Part C: Journal of Mechanical Engineering Science:

https://journals.sagepub.com/author-instructions/pic

Submission Guidelines: Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science: Sage Journals

Your report should demonstrate clarity of thought. Therefore, the reader must be told what to think and why. Don't leave the reader to complete a logical argument or draw his or her own conclusions. All points must be substantiated by one or more of the following

- logical argument based on established principles
- reference to published work
- your own results.

Don't believe everything you read. Before you cite a published result, read the paper and make sure you believe it. It often helps to summarise, in a brief phrase, how the author of a published paper reached a conclusion. If you think a published result is unreliable, don't ignore it but explain why.

2. The Reader

Assume your readers are graduate engineers. Remember that most normal people skipped some of the difficult bits in their degree syllabus or have forgotten things - so make allowances.

They want to know what you did, why you did it and what you found out. If they knew these things already, they wouldn't be reading your report! Explanations should tell your readers what you believe and provide all the logical steps which have led you to this belief.

Readers don't want to be bored or irritated by your report. They will be bored by:

- a tedious style
- repetition
- irrelevant material.

They may be irritated by:

- explanations which are not clear;
- an unattractive format;
- spelling mistakes;
- grammatical errors;
- diagrams which are not clear.

If you irritate or bore an examiner you risk failure.

3. Getting started

You should be thinking about your report throughout your project and discussing its contents with your supervisor. You should plan to start writing well before the Easter vacation. You should agree a time for your supervisor to read and comment on your first draft. This should be planned to allow yourself plenty of time (at least a week) to incorporate any comments into the final draft.

Discuss your report with your supervisor before you start. You may wish to write a plan before or following this discussion. The plan will probably be modified but should be a list of headings and sub-headings with a few notes on the contents of each. Choose a section which will be easy to write first, it probably won't be the Introduction!

4. Structure

4.1 Overview

Your report should consist of:

- A title page with your name
- An abstract
- The main body of the text divided into sections (and possibly sub-sections)
- Acknowledgements (if any)
- References
- Figures and Tables including their captions
- An appendix defining any symbols or abbreviations used in the text
- Any other short appendices that may be necessary (you may not need any).

4.2 Title page

The title should provide a clear statement of the topic of your report; you may wish to consider a short snappy phrase followed by a longer phrase which provides more information.

4.3 Abstract

The abstract should state the aim of your project, what you did (including techniques used), what you found out (including a summary of any numerical data), your conclusions and the implications of what you have done. It should be followed by a list of up to 5 keywords.

4.4 Main body of the text

This should contain no more than 5000 words and should be divided into sections. Sections may be divided into sub-sections but sub-sections should not be sub-divided.

For many projects it will typically be divided into the following sections:

- Introduction (usually without sub-headings)
- Methods
- Results
- Discussion
- Conclusions.

However, there may be reasons why you will to use different headings, especially for projects that are not based on experimental work or computer modelling.

4.5 Introduction

This should describe:

- the aims of the project
- the importance of the project (academic, industrial, commercial or other)
- any background material which is required to understand the report with references
- any previous work that is directly relevant to the project with references
- the structure of the report, if it is likely to be difficult to follow.

4.5 Methods

You should describe what you did in sufficient detail that someone could repeat your work in exactly the same way. Computer software, commercial equipment etc. should be specified exactly. Its name should be followed by the manufacturer or supplier, their town, county or state (if necessary and applicable) and country (if not UK). You may wish to state any statistical techniques you have used in this section.

4.6 Results

Results should be presented clearly, usually in tables or graphs. If you have a lot of results you may wish to provide a selection here and put the rest in an appendix; refer to the appendix when you present the results. Whenever possible, provide some measure of the precision of your results.

Numerical results should be presented to a reasonable number of significant figures. It is often better to write 1.00 x 10^3 rather than 1000, since the latter could imply that you are confident of a result to four significant figures when maybe you're not. Whenever possible give some measure of the precision and state what it is e.g. 1.31 ± 0.02 (mean $\pm$ SD).

Don't forget units. Use SI units and their correct abbreviations (e.g. "s" not "sec") whenever possible. Leave spaces (or place dots) between the parts of compound units; "ms" is the abbreviation for a millisecond but "m s" (or "m.s") is "metres x seconds". Don't put a full stop after the abbreviation for a unit. Don't add an "s" to make a unit plural.

4.7 Discussion

This should remind the reader of your aims, summarise your results and discuss their implications. You may wish to suggest future research which might be performed in this area. In particular, you will probably need to cover:

- the reliability of your results
- whether your results appear to be reasonable;
- how your findings relate to the work of others (give references)
- the applicability of your findings.

You may wish to include your Conclusions in this section.

4.8 Conclusions

You can state your conclusions in the Discussion section. However, they are often clearer if stated in a final short section.

4.9 Acknowledgements

You may wish to help people who have helped you with the project.

4.10 References

List all the sources of published information (including patents) which you have referred to in your thesis; don't include any extra publications. Don't make any mistakes - it's the one part of your report an examiner can easily check for carelessness. References should be cited and listed exactly as in the instructions (see IMechE Authors Guidelines).

Since there is a lot of unreliable information on the Web, references to the Web should be avoided. For general information cite a reliable text – not Wikipedia.

4.11 Tables

Tables are the best way of presenting lists of numerical data; they can also be used to list sources of materials etc. They should be clearly laid out with clear headings. It is usually best to include units for numbers in table headings.

Captions are placed above a table. A caption usually consists of a short phrase followed by a more detailed explanation, in sentences. You may wish to provide further explanation in numbered footnotes below the table, e.g. if you wish to comment on a particular result.

4.12 Figure captions

A caption usually consists of a short phrase followed by a more detailed explanation, in sentences. Consider grouping related figures together as parts a, b... of the same figure. When a figure is divided into parts, you should describe the parts at an early stage in the caption.

4.13 Figures

Figures should be clear, attractive and should not occupy excessive space. Keep diagrams simple. Colour pictures are preferable to black and white. Figures should be placed in the main part of the report as appropriate.

4.14 Appendices

You should define all abbreviations and symbols used in the report (including those in equations, figures and tables) in an Appendix. It is not necessary to define everyday abbreviations (like e.g.) but you should define technical abbreviations (like CFD). You do not need to define the standard abbreviations for SI units.

It is good practice to define abbreviations in the text when a subject is first mentioned e.g. "The stress distribution was calculated using finite element analysis (FEA)..."; after the abbreviation has been defined you should use it. Similarly, it is good practice to define symbols when they are first used e.g. "The applied torque, *T*, was represented by...".

You may use standard, everyday abbreviation (like e.g.) without definition but technical abbreviations (like FEA for finite element analysis) should be defined. By placing it in brackets after the first time the full wording is used. Readers can be forgetful so it may be a good idea to reintroduce an abbreviation in each chapter where it occurs. However, in some theses this may be overkill - so use your judgement. Sometimes abbreviations have full stops, sometimes they don't. It is becoming conventional to omit them in abbreviations of technical terms which consists of capital letters (like SD, CFD etc.) but to retain them in old favourites (e.g., i.e., etc.).

Further appendices may be used to cover anything which might disturb the flow of the report.

5. Style

5.1 General

Your report should be clearly expressed and free from grammatical or spelling errors. Try to make it interesting.

A report is more interesting if it is reasonably easy to understand and is not repetitive; this paragraph contains some hints. Explain why you are presenting information before you present it. Divide your text into paragraphs. Each paragraph should cover an idea. The first sentence of the paragraph should state the idea which is explained and justified in subsequent sentences. Make sure that each sentence conveys some useful information. Don't try to convey more than one idea in a sentence. Keep sentences short. Try to avoid paragraphs which are unusually short or long, except that a short paragraph may sometimes be useful for emphasis. Avoid using the same sentence to start consecutive sentences - unless you are confident that you can use repetition to give emphasis. Avoid using the same word several times in one sentence or in sentences which are next to each other. Avoid repeating ideas.

Use the past tense to describe what you did and the present tense to explain what is generally believed or accepted. Never write instructions. Thesis style should be more formal then this report. For example, the apostrophe should not be used to replace letters, e.g. write "do not" instead of "don't".

5.2 Words

Check your spelling. Mostly use English spelling but in some instances American spelling is more acceptable (e.g. write "programme" for a plan of work but "program" for computer software). Use your spell checker but, when in doubt, check in the Concise Oxford English Dictionary. Spelling of technical words should be checked in standard textbooks.

Some everyday words should be avoided. Don't use slang. Avoid jargon. Omit words which don't convey useful information - like "actually". Don't use complicated words or expressions when simple ones will do - write "now" not "at the present time". Some everyday words have specialist meanings in science (like "significant" and "complex"); it is better to use them only when their specialist meaning is intended.

Some words are frequently used incorrectly. "However" is not a conjunction, i.e. it should not be used like "and" to join parts of a sentence.

5.3 Grammar

It is sometimes debatable whether grammar is "correct" - sometimes you may disagree with your supervisor. Provided this disagreement is not simply a result of ignorance, write as you prefer - provided you can defend yourself against a hostile examiner.

A set of words which begins with a capital letter and ends in a full stop is a sentence - it must contain a verb.

5.4 Punctuation

You only need to read this section if your knowledge of punctuation is poor.

The apostrophe (') is never used to make a plural, i.e. write "in the 1960s..." not "in the 1960's. It is used to show possession (e.g. "Smith's idea") and to replace a missing letter. Don't use it the second way in your report. Note that "its" means "belonging to it"; "it's" means "it is" and shouldn't be used in your report. A comma is placed wherever you would pause in speech, e.g. between items in a list, after "however" etc. A semi-colon (;) is used to join two sentences which are closely related; the second does not then start with a capital letter. It can also be used to separate phrases in a list. A colon (:) is used to start a long list or to link two sentences which make a point by providing a contrast. ("United we stand: divided we fall.")

5.5 Copying, reporting and referencing

Copying or reporting the work of others without attributing the source is plagiarism. However, if you have developed ideas in collaboration with your supervisor you need not acknowledge this in your thesis, provided you acknowledge that he or she supervised your research in the "Acknowledgements". It is better to report other peoples' work in your own words, with an appropriate reference. If you decide to copy their wording, enclose it in inverted commas and don't forget the reference.

Sometimes it would be tedious to keep repeating a reference. You can overcome this by writing something like "Further details on all the statistical tests described in this section are given in standard textbooks..." in which you include the number of the reference you are citing. This sentence would appear in the first paragraph of the section.

Italics may be used for titles of books etc. or for foreign phrases. It is not common to italicise e.g., i.e. or etc. even though they are abbreviations of Latin phrases. However, it is common to italicise *et al*. Always be consistent; you can use a preface to defend yourself if you are concerned about a pedantic examiner. Don't use italic (or bold) type for emphasis.

5.6 Equations

Equations should occupy a separate line from the text; they should be indented and numbered sequentially. Use symbols to represent the quantities involved - not words; represent a quantity by a single symbol (although you may add subscripts). A quantity should be represented by a single symbol; use σ or s to represent standard deviation in an equation - not SD. Don't use the same symbol for two different things - so if you use σ to represent stress, don't use it to represent a standard deviation. If the symbol needs qualifying in some way - use a subscript e.g. initial and final times can be represented by t_o and t_f. Symbols in equations and text are usually given in italics.

6. Proof reading

It is your responsibility to proofread your report. Make sure that every sentence makes sense and conveys the intended meaning with no ambiguity. Check for spelling and grammatical errors. Watch out for inconsistencies.

APPENDIX B - Video Presentation Assessment Criteria

Percentage	Video Presentation (50 marks)	Q&A (50 Marks)
100%	A sophisticated presentation that explores the issues around reference material in an informed and perceptive way; enthusiastic and engaged delivery that makes the audience want to listen; uses very effective ways of conveying information, concepts and ideas; communicates difficult ideas in a clear and intelligible way. Perceptive and interesting responses to questions showing an excellent understanding and command of the subject area.	Perceptive responses to questions showing an excellent understanding and command of the subject area. The responses are highly reflective based on sound engineering judgement.
85%	An excellent presentation that explores the issues around reference material in an informed way; enthusiastic and engaged delivery that makes the audience want to listen; uses very effective ways of conveying information, concepts and ideas; communicates difficult ideas in a clear and intelligible way	Excellent responses to questions showing a detailed understanding and command of the subject area. The responses are often reflective based on sound engineering judgement.
78%	A very good presentation that explores the issues around reference material in an informed way; enthusiastic and engaged delivery with the audience; uses effective ways of conveying information, concepts and ideas; communicates difficult ideas in a clear way	Interesting responses to questions showing a very good understanding and command of the subject area.
75%		
72%		
68%	A carefully explained and clear presentation that conveys a good sense of the reference material used; clearly delivered presentation that shows an ability to engage directly with the audience; effective in conveying information, ideas and concepts to the audience.	Coherent and informed responses to questions.
65%		
62%		

58%	A coherent presentation but with some omissions or lack of clarity in the presentation of concepts and ideas; relatively clear delivery conveying some ideas and information with some audience engagement.	Clear but basic and limited responses to questions.
55%		
52%		
48%	Little defined structure or analysis; little or inadequate preparation; poorly delivered presentation with little engagement with the audience.	Responses to questions were incomplete and muddled.
45%		
42%		
35%	Failure to carry out the task assigned; rambling and unstructured content; no preparation; poor delivery and no engagement with the audience.	Answers to questions were irrelevant or mostly incorrect.
20%	Poor content and no structure; audience bored and uninterested.	Incoherent answers to questions.
0%	No presentation given.	No answers given.

APPENDIX C – Final Report Assessment Criteria

Percentage	Level of critical and deep understanding of chosen topic (20 Marks)	Quality of critical engagement with literature (20 Marks)	Choice and justification of research or design method (20 Marks)	Quality of analysis, and evaluation of findings or design (20 Marks)	Impact Statement (10 Marks)	Standard of presentation & structure of work (10 Marks)
100%	Demonstrates extensive critical and deep understanding of the topic in terms of depth and breadth. Extensive evidence of independent study. Draws extensively on and critiques latest developments in the subject area not covered on the programme. Level of critical and deep understanding goes considerably beyond what has been taught. Demonstrates extensive capacity for critical evaluation of new knowledge. Presents an extensive novel approach to the relevance and applicability of theory in the context of the research. The work is of publishable standard in a high factor impact journal. Error free.	Critical evaluation of the literature is extensively detailed and robust in both breadth and depth. Consolidates and extensively extends on knowledge in a novel/creative way. Key theories are critiqued extensively, challenged and/or interpreted in a novel way. Theories are synthesised extensively and related to the research. Literature used is considerably beyond the prescribed range (e.g. latest articles/policies/developments in subject area are drawn on). Of a standard of publishable work in a high impact factor journal. Error free.	Extensive evidence of innovative/creative research methodology. The justification of methods evidences a logic that goes considerably beyond the level taught. Methods are adapted to fit the research context in an imaginative and insightful way. The work is of a standard as observed in a published research article in a high impact factor journal. Error free.	Extensive analysis and evaluation of findings are critically sound in depth and breadth. There is extensive synthesis between research findings and the ideas presented in the literature review. Based on the research findings, key ideas in the literature are challenged in a way that goes markedly beyond the level at which the programme is taught. Conclusions made are warranted and extensively supported by the research. Several novel/creative insights are made. Standard is of publishable quality in a high impact factor journal. Error free.	An outstandingly clear description of the potential project impacts including Identification of all relevant economic/industrial, commercial, environmental, and societal project impact(s). High-level understanding and Identification of the nature and range of impact(s) including benefits, implementation & stakeholder considerations.	Written expressions are eloquent and coherent making it enjoyable and engaging to read throughout. All project milestones are met. Structure and navigation of work is elegant and flows seamlessly. References are all listed and cited correctly. Quality of a published paper in a high impact factor journal. Error free.
85%	Demonstrates high critical and deep understanding of the topic in terms of depth and breadth. Evidence of independent study. Draws on and critiques latest developments in the subject area not covered on the programme. Level of critical and deep understanding goes well beyond what has been taught. Demonstrates some capacity for critical evaluation of new knowledge. Presents a novel approach to the relevance and applicability of theory in the context of the research. The work is close to publishable standards.	Critical evaluation of the literature is detailed and robust in both breadth and depth. Consolidates and extends on knowledge in a novel/creative way. Key theories are critiqued, challenged and/or interpreted in a novel way. Theories are synthesised and related to the research. Literature used is beyond the prescribed range (e.g. latest articles/policies/developments in subject area are drawn on). Of a standard close to publishable work.	Evidence of innovative/creative research methodology. The justification of methods evidences a logic that goes beyond the level taught. Methods adapted to fit the research context. The work is close to a standard as observed in a published research article.	Analysis and evaluation of findings are critically sound in depth and breadth. There is synthesis between research findings and the ideas presented in the literature review. Based on the research findings, key ideas in the literature are challenged in a way that goes well beyond the level at which the programme is taught at. Conclusions made are warranted and supported by the research. A novel/creative insight is made. Standard is close to that required for publishable work.	A clear description of the majority of the potential project impacts. identifying most of relevant economic/industrial, commercial, environmental, and societal project impact(s). A very high-level understanding and Identification of the nature and range of impact(s) including benefits, implementation & stakeholder considerations.	Written expressions are coherent making it enjoyable and engaging to read throughout. Large majority of the project milestones are met. Structure and navigation of work flows seamlessly. References are all listed and cited correctly. Standard is close to that required for publishable work.

78% 75% 72%	Demonstrates critical and deep understanding of topic. Evidence of independent study. Draws on and critiques latest developments in the subject area not covered on the programme. Level of critical and deep understanding goes beyond what has been taught. Evidences the ability to relate concepts together and apply these to the context of the research.	Critical evaluation of the literature is robust and detailed. Consolidates and extends on knowledge. Key theories are critiqued, synthesised and related to the research. Literature selected is clearly beyond the prescribed range (e.g. latest articles in the subject area are drawn on).	Evidence of well-developed research methodology. The justification of methods evidences a logic that goes beyond the level taught. Methods adapted to the research context.	Analysis and evaluation of findings are critically sound. There is synthesis between research findings and the ideas presented in the literature review. This is achieved in a manner that goes beyond the level at which the programme is taught at. Conclusions made are warranted and supported by the research. Insights made go beyond expectations at this level.	Clearly outlining several examples of potential project impacts from amongst economic/industrial, commercial, environmental, and societal aspects. Excellent understanding and identification of the nature and range of impact(s) including benefits, implementation and/or stakeholder considerations.	Written expressions are clear and engaging, although contains the occasional grammar and spelling mistake. Majority of the project milestones are met. Structure and navigation of work flows seamlessly. References are all listed and cited correctly
68% 65% 62%	Demonstrates critical understanding of appropriate theoretical frameworks. Evidence of independent study. Extends beyond the readings covered on the modules. Some outside sources are critiqued. Evidences the ability to relate concepts together and apply these to the context of the research.	Critical evaluation of the literature is robust. Consolidates and extends on knowledge. Key theories are critiqued, synthesised and related to the research. The review draws on some outside sources.	Evidence of appropriate methodology. The justification of method choice is critical. Concepts are related together. Methods fit the research context.	Analysis and evaluation of findings are critically sound. There is synthesis between research findings and the ideas presented in the literature review. Conclusions made are warranted and on the whole supported by the research.	A good description of some of the project's potential impacts covering at least two from economic/industrial, commercial, environmental, and societal aspects. Evidence of a good understanding of the nature and/or range of impact(s) including benefits, perhaps with implementation of stakeholder considerations.	Written expressions are clear although contains some grammar and spelling mistakes. Several of the project milestones are met. Structure and navigation of work is clear and logical. References are cited correctly, with only minor errors.
58% 55% 52%	Demonstrates knowledge of appropriate theoretical frameworks. Evidence of independent study, although work is mainly, though not completely, centred on set sources. Level of understanding is balanced towards the descriptive, although some critical insights are made. Meaningful synthesis of theories and its application to the research is limited.	Review of the literature is balanced and descriptive. There is some evidence of the ability to critique, synthesise and apply key theories to the research. Consolidates and extends on knowledge to some extent. The review may draw on some outside sources. However, the review is reliant on the prescribed range from the module.	Appropriate methods selected. Justification is lacking in substance. Mainly descriptive, however there is some evidence of critical engagement. Methods fit the research context.	Discussion is on the whole balanced towards the descriptive rather than the critical. Limited meaningful synthesis is made between findings and literature. Conclusions made are sound although not fully justified by the research as a whole.	A moderate description of some of the project's potential impacts covering at least two from economic/industrial, commercial, environmental, and societal aspects. Some evidence of a limited understanding of the nature and/or range of impact(s) including benefits.	Written expressions are clear. Work contains some grammar and spelling mistakes. Several of the project milestones are met. Structure of work lacks coherency in some places. However, this does not greatly impact on navigation. References are cited, but some are incomplete and/or contain errors.

48% 45% 42%	Demonstrates an awareness of theoretical frameworks, but level of understanding is mainly descriptive. The work lacks meaningful synthesis of theories and its application to the research. Evidence of independent study is limited. Sources drawn on are constrained to the readings covered on the programme.	Review of the literature is descriptive. Meaningful synthesis and application of the key theories to the research is absent. There is evidence of some ability to evaluate core readings. The literature used is limited to the prescribed range from the programme.	Methods selected are somewhat appropriate. Justification for choice of methods is descriptive and knowledge is limited to basic facts and concepts.	Discussion of findings is descriptive. No meaningful synthesis is made between findings and the literature review. Basic conclusions are drawn which are not fully justified by the research as a whole.	A basic description of some of the project's potential impacts covering at least one from economic/industrial, commercial, environmental, and societal aspects. Some evidence of basic understanding of the nature and/or range of impact(s) including benefits.	Written expressions are competent but lacks clarity in places due to minor issues with sentence construction. Work contains some spelling and grammar mistakes. A limited number of the project milestones are met. Structure of work lacks coherency. This makes navigating the work difficult in places. Referencing is incomplete and/or incorrect.
35%	Demonstrates an awareness of some of the key theoretical frameworks with some simple connections made. Significant theories are not covered and/or major mistakes in interpretation made.	Review is descriptive. Work fails to address some aspects of the brief. Needs to provide some evaluation (e.g. note the strengths and weaknesses) of the core literature to achieve a pass standard.	Inappropriate choice of methods. Poorly argued/illogical justification given for choice of methods.	Discussion of findings is limited and descriptive (e.g. does not reflect on/describe all the research findings). Misses the point. No meaningful synthesis is made between findings and literature review. Conclusions are not coherent/irrelevant.	A mostly unclear description of some of the project's potential impacts with little evidence of understanding of the nature and/or range of impact(s) including benefits.	Written expressions lack clarity. Key points are not clearly articulated. Work contains frequent spelling mistakes. There are issues with sentence construction. The structure of work is not coherent. Referencing is mostly incomplete and/or incorrect. The majority of the project milestones are not met.
20%	Does not demonstrate an understanding of key theoretical frameworks. Level of understanding is at the word level with facts being reproduced in a disjointed or decontextualised manner.	Inappropriate choice of methods. Poorly argued/illogical justification given for choice of methods.	Inappropriate choice of methods made or choice of methods is unclear. No justification given for choice of methods. Facts are reproduced in a disjointed or decontextualized manner.	No/little attempt is made to engage with or describe the findings. Conclusions are absent.	Impact statement lacks clarity throughout with little or no attempt to provide its nature or range.	Written expressions lack clarity. No attempt is made to articulate key points. Work contains frequent spelling mistakes. There are issues with sentence construction. The structure of work is not coherent. No attempt at referencing is made or most references are typically missing. No project milestones are met.
0%	No critical understanding of chosen topic.	No critical engagement with literature.	No justification of research methods or design methods.	No attempt is made to engage with or describe the findings. Conclusions are absent.	No impact statement given.	No evidence of a structure to the report. No attempt at referencing is made. Report is unreadable. No project milestones are met.

APPENDIX D

Summary Project Meeting Form

Instructions:
Complete Sections 1, 2 and 3 and submit to Canvas shortly AFTER your meeting with your supervisor.
In total, you will need to submit 5 Project Meeting forms, each with a separate Canvas deadline.

Section 1

Student's Full Name:		**Student's ID:**	
Supervisor's Name:		**Meeting Date:**	

Section 2

Project Title:	

In the space below write a summary of the outcome of the previous meeting:
In the space below write a summary of the progress on your project since last meeting:

Section 3

Sign *or* print your name:	
Date of next meeting:	

By submitting this form to Canvas you confirm the accuracy of the information you have provided above.